



Response to the OSTP Request for Information

Development of an Artificial Intelligence Action Plan

The [Society for Industrial and Organizational Psychology](#) (SIOP) appreciates the opportunity to provide the below response to the White House Office of Science and Technology Policy (OSTP) Request for Information (RFI) on the development of an artificial intelligence action plan. SIOP is the premier professional association for the science and practice of industrial and organizational (I-O) psychology. We are a 9,000+ member organization based in the US. While an independent organization with its own governance, SIOP also represents Division 14 of the American Psychological Association and is an organizational affiliate of the Association for Psychological Science.

I-O psychology is a dynamic and growing field that addresses workplace issues at the individual, team, and organizational levels. I-O psychologists apply research that improves the well-being and performance of people and the organizations that employ them. This involves everything from workforce planning, selecting employees, and developing leaders to studying job attitudes and motivation, implementing work teams, and facilitating organizational change. Particularly relevant to this RFI, I-O psychologists are also rigorously trained in the development and evaluation of tests, assessments, and other selection procedures that are used to make hiring and promotion decisions. Using their expertise, I-O psychologists have been working to improve the accuracy and effectiveness of selection procedures for decades.

SIOP's Subcommittee on Artificial Intelligence (AI)-Based Assessments was initially launched in October 2021 as a task force to address issues related to the development and implementation of AI for assessing and hiring talent and to increase awareness of scientific research on this topic. The current subcommittee comprises SIOP members with expertise in a broad range of related areas such as employee selection and assessment, psychometrics and measurement bias, and the use of AI-based technologies in the workplace. I-O psychologists with this AI expertise are already employed at Fortune 50 companies and leading technology firms. Many in our profession are thus well-suited to help address organizational and workforce issues related to OSTP development of an artificial intelligence action plan.

Given the increasing prevalence and sophistication of AI-based technologies and systems in the modern workplace and the need to ensure greater innovation and effectiveness in all aspects of the American workforce, **SIOP encourages the OSTP national artificial intelligence action plan to incorporate plans for (a) improving AI-based hiring and promotion decisions and (b) accelerating a competitive AI-enabled workforce that is capable of working effectively with AI-based tools.**

Improving AI-Based Hiring Decisions

Hiring decisions are made many thousands of times each day, and many organizations, including federal agencies, are looking for ways to make these decisions more accurately, efficiently, and cost-effectively to remain competitive in a global economy. To this end, AI is being increasingly incorporated into the employment screening of job candidates. In this context, a broad range of AI technologies and statistical techniques seek to identify patterns in candidate information that are predictive of future job performance. At the same time, there have been increasing calls for greater legal scrutiny of AI-based hiring practices, reflecting concerns over privacy, fairness, lack of transparency, and the accuracy of AI predictions.

For many decades, SIOP and the I-O psychologists it represents have developed and implemented professional standards for creating effective and legally defensible employment tests. SIOP asserts that AI-based assessments used to make hiring and promotion decisions require the same level of scrutiny and should meet the same standards that traditional employment tests have been subjected to for decades. In other words, **AI-based assessments should still be required to meet the standards for traditional hiring and assessment procedures, even if different methods are used to meet those standards.**

To guide professionals and inform those responsible for staffing in organizations, the Society for Industrial and Organizational Psychology (SIOP) has published the [*Principles for the Validation and Use of Personnel Selection Procedures*](#) (SIOP, 2018; hereafter referred to as the *Principles*), which is updated regularly to reflect current scientific research and best practices in hiring and promotion. This document summarizes the fundamental requirements for selection procedures that should guide the evaluation of assessments. Importantly, these professional guidelines are applicable to *all selection procedures*, including technology-based hiring and promotion procedures that incorporate AI, machine learning, and other novel assessment techniques (e.g., game-based assessments, evaluation of voice and facial characteristics).

Although the *Principles* are applicable to *all personnel selection procedures*, the development and use of AI-based procedures presents unique challenges. To address the unique challenges and considerations that arise when using AI-based assessments and hiring procedures, SIOP released the [*Considerations and Recommendations for the Validation and Use of AI-Based Assessments for Employee Selection*](#) (SIOP, 2023). This document provides recommendations rather than mandates for the use of AI and notes that the professional judgment of experts on hiring and assessment can provide essential additional guidance on the applicability and necessity of the specific recommendations provided.

The [*recommendations*](#) released by SIOP on this topic specify four key criteria for evaluating AI-based assessments:



1. AI-based assessments should produce scores that predict future job performance or other relevant outcomes accurately.
2. AI-based assessments should produce consistent scores (e.g., upon re-assessment) that reflect job-related characteristics.
3. AI-based assessments should produce scores that minimize job-irrelevant characteristics and are otherwise considered fair and unbiased for all job applicants.
4. All steps and decisions related to the development and scoring of AI-based assessments should be documented for verification and auditing.

To avoid being unnecessarily burdensome, these four key criteria are intended to represent the minimal requirements necessary to justify the use of AI-based assessments for hiring and promotion decisions. Most importantly, following these recommendations will not hamper AI innovation. Instead, they suggest that innovative AI technologies and approaches should be rigorously evaluated *before* they are used to make high-stakes employment decisions to ensure they are accurate, effective, and work as intended.

Accelerating the AI-Enabled Workforce

Understanding how AI and automation can accelerate the capabilities of the American workforce has never been more critical. AI is rapidly transforming organizations and the goods and services they produce. Employees are increasingly coworking with robots and AI, and AI is becoming deeply embedded into their technological and physical workspaces. Such AI workforce trends demand that employees reskill and adapt at scale.

I-O psychologists have deep expertise and experience in helping US organizations and the workforce prepare for AI and other major technological shifts. Assessing job demands and developing responsive employee training programs using AI technologies are necessary to inform future efforts to reskill employees. As jobs are transformed or replaced, it is also critical to understand how American workers can seek out new opportunities and otherwise adapt successfully in today's labor market for the US economy to remain a dominant force. Assessing, predicting, interpreting, and improving employee and work team interactions with automation and other new technologies is one key example of core I-O training that contributes to organizational success. I-O psychologists also have expertise in developing, implementing, and evaluating effective reskilling strategies, including identifying current skill needs and projecting the skills that will be important for the current and future job market. Efficient job retraining can address the need for lifelong skills education, keep older workers in the workforce, and combat talent shortages in areas critical for innovation and societal well-being.

In the action plan, SIOP encourages an increased focus on bolstering the AI-enabled workforce, including understanding the changing skills that will be needed. SIOP additionally encourages OSTP to include research on improving and disseminating strategies for effective upskilling and



job training, particularly workplace-based training that allows employees to continue working while expanding their job capabilities and skills. Finally, SIOP recommends that OSTP include a focus on anticipating and responding to worker mobility needs given the increased use of AI and other automation in the workplace to ensure that workers will not merely remain productive but maximally effective in their contributions to a world-leading US economy.

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Expert Contacts

SIOP welcomes the opportunity to submit these comments and provide further expertise and insight as OSTP seeks to develop the National AI Action Plan. Please reach out to the following SIOP issue experts with additional comments or questions:

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References

Society for Industrial and Organizational Psychology. (2018). *Principles for the Validation and Use of Personnel Selection Procedures* (5th ed.). Retrieved from: <https://www.apa.org/ed/accreditation/about/policies/personnel-selection-procedures.pdf>.



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